Seminar 8 – 25.11.2021. – 312CD. – Algaed.

Spații euclidiene reale

Def. $(V,+,\cdot)$ sp.vect / $\mathbb{R}$.

$(\cdot,\cdot): V\times V \to \mathbb{R}$ s.n. produs scalar dc.

1) $(u,v)=(v,u)$; $\forall u,v\in V$.

2) $\langle u+v, w\rangle = \langle u,w\rangle + \langle v,w\rangle$

3) $\langle \lambda u, v\rangle = \lambda\cdot\langle u,v\rangle$; $\forall \lambda\in\mathbb{R}$

4) $\langle u,u\rangle \geq 0$; $\forall u\in V$.
$\langle u,u\rangle = 0 \Leftrightarrow u = 0_V$.

Exp. 1) $\mathbb{R}^n/\mathbb{R}$ $\quad x=(x_1,\dots,x_n)$
$y=(y_1,\dots,y_n)$.
$\langle x,y\rangle = \sum_{i=1}^{n} x_i\cdot y_i = x_1y_1+\dots+x_ny_n$ e produs scalar

2) $V=\{f:[a,b]\to\mathbb{R} \mid f \text{ cont}\}$
$\langle f,g\rangle = \int_a^b f(t)g(t)\,dt \to$ produs scalar.

3) Ar.că fct $\langle\cdot,\cdot\rangle: \mathbb{R}^2\times\mathbb{R}^2\to\mathbb{R}$.
$\langle x,y\rangle = x_1y_1+x_1y_2+x_2y_1+2x_2y_2$ e produs scalar.
$x=(x_1,x_2)$; $y=(y_1,y_2)$ pe $V=\mathbb{R}^2$.

① $\langle x,y\rangle = x_1y_1+x_1y_2+x_2y_1+2x_2y_2$ } $\Rightarrow \langle x,y\rangle = \langle y,x\rangle$.
$\langle y,x\rangle = y_1x_1+y_1x_2+y_2x_1+2y_2x_2$ =

② $\langle x+y,z\rangle = (x_1+y_1)\cdot z_1 + (x_1+y_1)\cdot z_2 + (x_2+y_2)\cdot z_1 +$
$+2\cdot(x_2+y_2)\cdot z_2 = (x_1z_1+x_1z_2+x_2z_1+2x_2z_2) +$
$+(y_1z_1+y_1z_2+y_2z_1+2y_2z_2) = \langle x,z\rangle + \langle y,z\rangle$.

③ $\langle \lambda x,y\rangle = (\lambda x_1)y_1 + (\lambda x_1)y_2 + \lambda x_2\cdot y_1 + 2\lambda x_2\cdot y_2 =$
$= \lambda[x_1y_1+x_1y_2+x_2y_1+2x_2y_2] = \lambda\cdot\langle x,y\rangle$.

④ $\langle x,x\rangle = x_1^2+x_1x_2+x_2x_1+2x_2^2 =$
$= (x_1^2+2x_1x_2+x_2^2)+x_2^2 = (x_1+x_2)^2+x_2^2 \geq 0$.

$\langle x,x\rangle = 0 \Leftrightarrow (x_1+x_2)^2+x_2^2=0 \Leftrightarrow \begin{cases} x_1+x_2=0 \\ x_2=0 \end{cases} \Leftrightarrow x=(0,0)$

1,..,4 $\Rightarrow \langle\cdot,\cdot\rangle \Rightarrow$ pr. scalar.

Exp. $\mathbb{R}^2$. $x=(x_1,x_2)$. $y=(y_1,y_2)$. $\langle x,y\rangle = x_1y_1 + x_2y_1$.
nu e produs scalar.

$\langle y,x\rangle = y_1x_1 + y_2x_1 \neq x_1y_1 + x_2y_1 = \langle x,y\rangle$.
$y_2x_1 \neq x_2y_1$

$x=(2,3)$; $y=(4,5) \Rightarrow y_2x_1 = 10$; $x_2y_1 = 12$. ▨

Ortogonalitate $(V,+,\cdot)$; $\langle\cdot,\cdot\rangle$) sp. euclidian real.

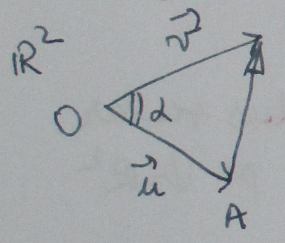
- $u \perp v \Leftrightarrow \langle u,v\rangle = 0$.
- **Norma** $\boxed{\|u\| = \sqrt{\langle u,u\rangle}}$.

i) $\|u\| \geq 0$.
ii) $\|u\| = 0 \Leftrightarrow u = 0$.
$\|\lambda\cdot u\| = \sqrt{\langle \lambda u, \lambda u\rangle} = \sqrt{\lambda^2\langle u,u\rangle} = |\lambda|\cdot\|u\|$.

$\|\cdot\| : V \to \mathbb{R}_+$

- $\boxed{\cos \sphericalangle(u,v) = \dfrac{\langle u,v\rangle}{\|u\|\cdot\|v\|}}$.

- **Legătura cu cosinusul din liceu**



$\vec{OA} = \vec{u}$
$\vec{OB} = \vec{v}$
$\vec{AB} = \vec{v} - \vec{u} \Rightarrow \|\vec{AB}\|^2 = \|\vec{v}-\vec{u}\|^2 = \langle \vec{v}-u, \vec{v}-\vec{u}\rangle$

ΔAOB: $AB^2 = OA^2 + OB^2 - 2OA\cdot OB\cdot\cos\alpha$. (1).
T. cosinusului (Pitagora generalizată)

MS $= AB^2 = \|\vec{AB}\|^2 = \langle \vec{v}-\vec{u}, \vec{v}-\vec{u}\rangle = \langle \vec{v},\vec{v}\rangle - \langle \vec{u},\vec{v}\rangle - \langle \vec{v},\vec{u}\rangle + \langle \vec{u},\vec{u}\rangle$
$= \|\vec{v}\|^2 + \|\vec{u}\|^2 - 2\langle u,v\rangle$. (2)

MD $= OA^2 + OB^2 - 2\cdot OA\cdot OB\cdot\cos\alpha = \|\vec{OA}\|^2 + \|\vec{OB}\|^2 - 2\|\vec{OA}\|\cdot\|\vec{OB}\|\cdot\cos\alpha$
$= \|\vec{u}\|^2 + \|\vec{v}\|^2 - 2\cdot\|\vec{u}\|\cdot\|\vec{v}\|\cdot\cos\alpha$. (3)

(1), (2), (3) $\Rightarrow \langle u,v\rangle = \|u\|\cdot\|v\|\cdot\cos\alpha$.

$\boxed{\cos\alpha = \dfrac{\langle u,v\rangle}{\|u\|\cdot\|v\|}}$, ▨

Def: $B=\{v_1,\dots,v_n\}$ **bază ortogonală** pt V dc $\langle$ bază $v_i \perp v_j$; $\forall i\neq j$.

Def. $B=\{v_1,\dots,v_n\}$ **bază ortonormată** pt V dc $\langle$ bază ortogonală $\|v_i\|=1$; $\forall i$.

Ex2. Tm sp. euclidian $\mathbb{R}^3$

$B=\{e_1=(1,0,1);\ e_2=(2,1,-3),\ e_3=(-1,1,0)\}$ bază.

a) ? B bază ortogonală.

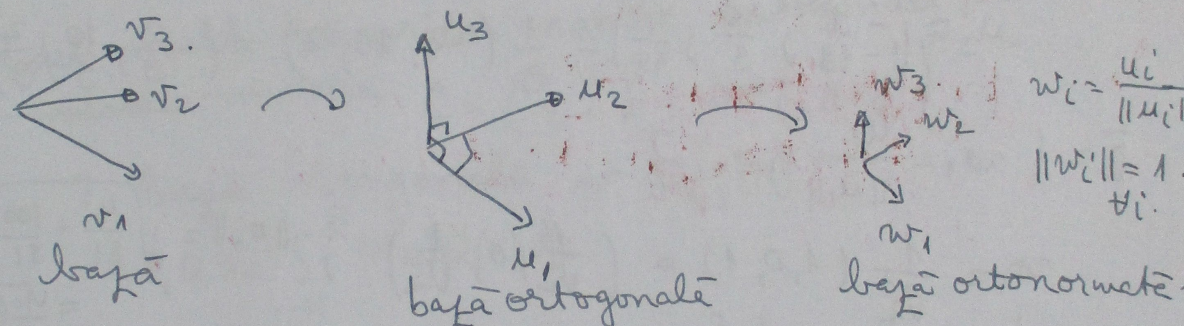
b) Constr o bază ortonormată pe $\mathbb{R}^3$ pornind de la baza B.

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a) $\langle e_1,e_2\rangle = 1\cdot 2+0\cdot 1+1\cdot(-3)=-1\neq 0 \Rightarrow$ B nu e bază ortogonală

Algoritmul Gram-Schmidt.

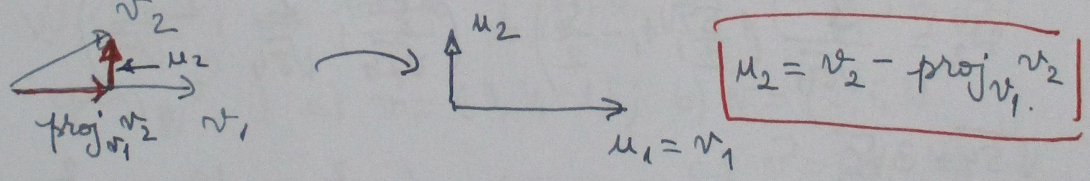
$\mathbb{R}^3$



$w_i=\frac{u_i}{\|u_i\|}$.

$\|w_i\|=1$. $\forall i$.

$\mathbb{R}^2$



$$u_2 = v_2 - \text{proj}_{v_1} v_2.$$

$\text{proj}_{v_1} v_2 = \alpha\cdot v_1$.

$u_2 = v_2 - \alpha v_1$

$\Rightarrow u_2 \perp v_1 \Rightarrow \langle u_2, v_1\rangle = 0$.

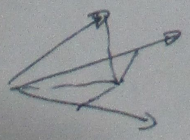
$\langle v_2-\alpha v_1, v_1\rangle = 0$.

$\langle v_2, v_1\rangle - \alpha\langle v_1, v_1\rangle = 0$

$$\alpha = \frac{\langle v_2, v_1\rangle}{\langle v_1, v_1\rangle}.$$

$$\text{proj}_{v_1} v_2 = \frac{\langle v_2, v_1\rangle}{\langle v_1, v_1\rangle}\cdot v_1$$

$\mathbb{R}^3$ $\{v_1,v_2,v_3\}$.



$$\begin{cases} u_1 = v_1 \\ u_2 = v_2 - \frac{\langle v_2, u_1\rangle}{\langle u_1, u_1\rangle}\cdot u_1 \\ u_3 = v_3 - \frac{\langle v_3, u_1\rangle}{\langle u_1, u_1\rangle}\cdot u_1 - \frac{\langle v_3, u_2\rangle}{\langle u_2, u_2\rangle}\cdot u_2 \end{cases}$$

$\rightarrow w_i = \frac{u_i}{\|u_i\|}$.

$v_1=(1,0,1)$; $v_2=(2,1,-3)$; $v_3=(-1,1,0)$.

$u_1=v_1=(1,0,1)$

$u_2=v_2-\frac{\langle v_2,u_1\rangle}{\langle u_1,u_1\rangle}\cdot u_1=(2,1,-3)-\frac{2+0-3}{2}(1,0,1)\Rightarrow$

$u_2=(2,1,-3)+\left(\frac{1}{2},0,\frac{1}{2}\right)=\left(\frac{5}{2},1,-\frac{5}{2}\right)$.

$u_3=v_3-\frac{\langle v_3,u_1\rangle}{\langle u_1,u_1\rangle}u_1-\frac{\langle v_3,u_2\rangle}{\langle u_2,u_2\rangle}u_2=(-1,1,0)-\frac{-1}{2}(1,0,1)-\frac{-\frac{3}{2}}{\frac{54}{4}}u_2$

$\langle v_3,u_1\rangle=-1$; $\langle v_3,u_2\rangle=-\frac{5}{2}+1=-\frac{3}{2}$

$\langle u_2,u_2\rangle=\frac{25}{4}+1+\frac{25}{4}=\frac{54}{4}$

$u_3=(-1,1,0)+\left(\frac{1}{2},0,\frac{1}{2}\right)+\frac{1}{9}\left(\frac{5}{2},1,-\frac{5}{2}\right)=\left(-\frac{1}{2},1,\frac{1}{2}\right)+\left(\frac{5}{18},\frac{1}{9},-\frac{5}{18}\right)$

$u_3=\left(-\frac{4}{18},\frac{10}{9},\frac{4}{18}\right)=\frac{1}{9}(-2,10,2)=\left(-\frac{2}{9},\frac{10}{9},\frac{2}{9}\right)$.

$B=\left\{w_1=\frac{u_1}{\|u_1\|},\dots\right\}$

$w_1=\frac{1}{\sqrt{2}}(1,0,1)=\left(\frac{1}{\sqrt{2}},0,\frac{1}{\sqrt{2}}\right)$. ; $\|u_3\|=\sqrt{\frac{4}{81}+\frac{100}{81}+\frac{4}{81}}=\frac{\sqrt{108}}{9}$.

$w_2=\frac{2}{\sqrt{54}}\left(\frac{5}{2},1,-\frac{5}{2}\right)=\left(\frac{5}{\sqrt{54}},\frac{2}{\sqrt{54}},-\frac{5}{\sqrt{54}}\right)$.

$\sqrt{54}=3\sqrt{6}$.

$w_3=\frac{9}{\sqrt{108}}\left(-\frac{2}{9},\frac{10}{9},\frac{2}{9}\right)=\left(-\frac{2}{\sqrt{108}},\frac{10}{\sqrt{108}},\frac{2}{\sqrt{108}}\right)$.

$108\,|\,9$
12

$\sqrt{108}=6\sqrt{3}$.

$w_3=\left(-\frac{1}{3\sqrt{3}},\frac{5}{3\sqrt{3}},\frac{1}{3\sqrt{3}}\right)$.

$\boxed{w_3=\left(-\frac{\sqrt{3}}{9},\frac{5\sqrt{3}}{9},\frac{\sqrt{3}}{9}\right)}$.

$\{w_1,w_2,w_3\}$. bază ortonormată.

Complementul ortogonal al unui subsp.

$S \subseteq V$ / $S^\perp = \{ v / v \perp u ; \forall u \in S \}$.
subsp

$S \oplus S^\perp = V$; $\dim S + \dim S^\perp = \dim V$.

Ex 3 . $S = \{ x \in \mathbb{R}^3 ; x = (x_1, x_2, x_3) ; x_1 + x_2 = 0 \}$.

a) $S^\perp = ?$.

b) descomp. vect $v = (1,4,5)$ după cele 2 subsp.

Ex 4 . $\mathbb{R}^4$; $U = \{ x = (x_1, \dots, x_4) ; x_1 + x_2 + x_3 + x_4 = 0 .$
$x_1 - x_2 + x_3 - x_4 = 0 \}$.

$U^\perp = ?$

Ex 5 . $V = M_2(\mathbb{R})$; $\langle A, B \rangle = tr(A^t B)$. produs scalar .

? dut mult. versorilor matrice ortogonali pe vectorii-matrice $C = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$; $D = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}$.

Ex 6 . ? bază ortogonală pt $\mathbb{R}_{\leq 2}[X]$

$\langle P, Q \rangle = \int_{-\pi}^{\pi} P(x) Q(x) dx$.

Ex 7 . $W \subseteq \mathbb{R}^3$ cu baza $\left(\frac{1}{\sqrt{5}}, 0, \frac{2}{\sqrt{5}} \right)$; $\left(-\frac{2}{\sqrt{5}}, 0, \frac{1}{\sqrt{5}} \right)$.

$\text{proj}_W \vec{v}$; a) $\vec{v} = (3,4,-1)$; b) $\vec{v} = (2,1,3)$.

c) $\vec{v} = (8,0,4)$.

Ex 3 . $S = \{ x = (x_1, x_2, x_3) \in \mathbb{R}^3 / x_1 + x_2 = 0 \}$. a) $S^\perp = ?$.

$S = Sp\{v_1, \dots, v_n\}$.

$S^\perp = \{ u \in V / u \perp v_i ; i = \overline{1,n} \}$.

? bază pt S .

$x = (x_1, x_2, x_3) \in S \Rightarrow x_2 = -x_1 \Rightarrow x = (x_1, -x_1, x_3)$

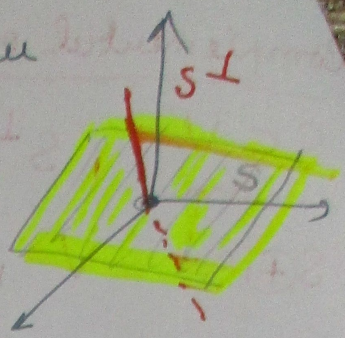
$x = x_1 \underbrace{(1,-1,0)}_{v_1} + x_3 \underbrace{(0,0,1)}_{v_2}$. $\Rightarrow S = Sp\{v_1, v_2\}$. $\Rightarrow \{v_1, v_2\}$ bază pt S.

v_1, v_2 liniind.

$T = \begin{pmatrix} 1 & 0 \\ -1 & 0 \\ 0 & 1 \end{pmatrix}$

$\dim S = 2$.

$S = Sp\{v_1 = (1,-1,0);\ v_2 = (0,0,1)\}$. plan



$\dim S + \dim S^\perp = \dim \mathbb{R}^3$.

$\dim S^\perp = 1$

$S^\perp = \{u \in \mathbb{R}^3 / u \perp v_1;\ u \perp v_2\}$.

$u = (a,b,c) \in S^\perp \Rightarrow \begin{matrix} \langle u, v_1\rangle = 0 \\ \langle u, v_2\rangle = 0 \end{matrix} \Rightarrow \begin{cases} a - b = 0 \\ c = 0 \end{cases} \Rightarrow a = b$

$u = (a,a,0) = a(1,1,0)$.

$S^\perp = \{a\underbrace{(1,1,0)}_{v_3}\} = Sp\{v_3\}$; $\{v_3\}$ bază pt $S^\perp$.

$\dim S^\perp = 1$.

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b) $v = (1,4,5) \Rightarrow$ descompunem după S și $S^\perp$.

$v = \underbrace{u}_{\in S} + \underbrace{w}_{\in S^\perp}$.

$S \oplus S^\perp = V$.

$S \cap S^\perp = \{0_V\}$

R_1: $\{v_1, v_2, v_3\}$ bază pt $\mathbb{R}^3$

$v = \alpha v_1 + \beta v_2 + \gamma v_3 \Rightarrow \begin{pmatrix}1\\4\\5\end{pmatrix} = \underbrace{\alpha\begin{pmatrix}1\\-1\\0\end{pmatrix} + \beta\begin{pmatrix}0\\0\\1\end{pmatrix}}_{u} + \gamma\begin{pmatrix}1\\1\\0\end{pmatrix}$.

$\begin{cases} 1 = \alpha + \gamma \\ 4 = -\alpha + \gamma \\ 5 = \beta \end{cases} \Rightarrow \gamma = \frac{5}{2} \Rightarrow \alpha = -\frac{3}{2}$.

$u = \alpha v_1 + \beta v_2 = (-\frac{3}{2}, \frac{3}{2}, 0) + \frac{5}{2}(0,0,1)$

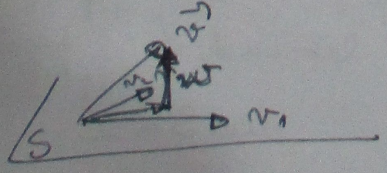
$u = (-\frac{3}{2}, \frac{3}{2}, \frac{5}{2}) \in S$.

$w = \gamma \cdot v_3 = \frac{5}{2}\begin{pmatrix}1\\1\\0\end{pmatrix}^t = (\frac{5}{2}, \frac{5}{2}, 0)$.

$v = \underbrace{(-\frac{3}{2}, \frac{3}{2}, 5)}_{\in S} + \underbrace{(\frac{5}{2}, \frac{5}{2}, 0)}_{\in S^\perp} = (1,4,5)$.

R_2 $S = Sp\{v_1, v_2\}$; $\langle v_1, v_2\rangle = 0 \Rightarrow \{v_1, v_2\}$ bază ortogonală

$v_1 = (1,-1,0)$; $v_2 = (0,0,1)$.



$proj_S v = proj_{v_1} \vec{v} + proj_{v_2} \vec{v}$.

$v = \underbrace{u}_{\in S} + \underbrace{w}_{\in S^\perp}$; $\boxed{\begin{matrix} u = proj_S v \\ w = v - proj_S v \end{matrix}}$ (!!!...)

Ex 5. $\exists A \in M_2(\mathbb{R})$ s.t. $\begin{cases} \langle A, C \rangle = 0 \\ \langle A, D \rangle = 0 \\ \|A\| = \sqrt{\langle A, A \rangle} = 1 \end{cases}$

$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow \langle A, C \rangle = tr(A^t C) = tr\left(\begin{pmatrix} a & c \\ b & d \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}\right)$

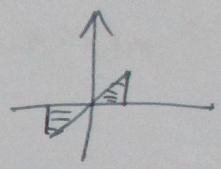
Ex 6. ? bază ortogonală pt $\mathbb{R}_{\leq 2}[X]$.

$\{1, X, X^2\}$ bază pt $\mathbb{R}_{\leq 2}[X]$. (G-S) bază ortog.

$P_1\ P_2\ P_3 \longrightarrow \{Q_1, Q_2, Q_3\}$. $\langle Q_i, Q_j \rangle = 0$.

$Q_1 = P_1 = 1$.

$Q_2 = P_2 - \frac{\langle P_2, Q_1 \rangle}{\langle Q_1, Q_1 \rangle} \cdot Q_1 = P_2 = X$.



$\langle P_2, Q_1 \rangle = \int_{-\pi}^{\pi} P_2(x) \cdot Q_1(x)\,dx = \int_{-\pi}^{\pi} x \cdot 1\,dx = 0$.

$Q_3 = P_3 - \cdot \quad \cdot\cdot -$